

# Genome Biology 投稿系统填写清单

版本锚点: `paralog-sl-predictor @ d671ea9` (manuscript.pdf 29 页 / supplementary.pdf 17 页 / cover\_letter.pdf 3 页)  
配套文件: highlights 与图形摘要说明文字见 `submission_texts.md` (已含字符数核对) 使用前请完成第 0 节的提交前核对;【待补】为待补项。

## 0. 提交前最终核对 (投稿当天执行)

```
cd paralog_sl_predictor && bash verify_all.sh
# 期望输出: 165/165 claims match; 31 passed; ALL CHECKS PASSED
```

- ☐ `verify_all.sh` 全绿
- ☐ 三份 PDF 为最新编译 (manuscript 29 页 / supplementary 17 页 / cover letter 3 页)
- ☐ 作者名单与邮箱最终确认 (见第 2 节【待补】)
- ☐ 摘要版本最终选定 (见第 3 节【注意】当前 273 词超限, 须用压缩版)

## 1. 期刊与文章类型

| 系统字段         | 填写内容                                                 |
|--------------|------------------------------------------------------|
| Journal      | Genome Biology                                       |
| Article type | Research                                             |
| Section      | 按系统列表选最接近者 (Cancer genomics / Computational biology) |

## 2. 标题 (Title)

Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types

## 3. 摘要 (Abstract, 结构化, 系统上限 250 词)

【注意】文稿当前摘要为 273 词, 超过系统 250 词硬限制。以下为 236 词 (含小节标签 239 词) 压缩建议稿, 所有数字与正文一致、限定语全部保留; 如采纳, 请同步替换 `manuscript.tex` 摘要后重编 PDF (或仅在系统文本框使用压缩版, 二者择一, 建议前者保持一致):

**Background:** Paralog compensation is a specific mechanism of synthetic lethality (SL): when a tumor-suppressor driver gene is lost to mutation, its sequence-similar paralog can become conditionally essential. Systematic prioritization of paralog-SL candidates remains limited because existing computational methods are black boxes that cannot explain their predictions, hindering targeted validation.

**Results:** We introduce Delta Dependency (DD), a single interpretable metric measuring the dependency shift on a paralog between driver-mutant and wild-type cell lines in DepMap (1,208 lines, 23 solid tumor types). In the pre-specified primary evaluation, a lineage-level benchmark on twelve curated, evidence-tiered paralog-SL pairs, DD achieved AUROC = 0.676 (0.725 excluding two pairs with DepMap-derived labels; 1.000 on the Tier A/B external benchmark, two of five pairs lineage-evaluable). In pair-level evaluation on the same 72 test pairs, the interpretable composite score reached AUROC = 0.831, statistically indistinguishable from the best multi-feature classifier (SVM-RBF, 0.843) under leave-one-pair-out cross-validation. Exploratory analyses across seven CPTAC cohorts (n = 771) detected protein-level paralog co-variation invisible at the RNA level. Dependency-window scoring prioritized ARID1A->ARID1B (DWS = 2.82) as the leading selective candidate, suggesting a hypothesis-generating biomarker strategy centered on truncating ARID1A mutations.

**Conclusions:** DD offers a mechanistically transparent alternative to black-box SL prediction: each nomination traces to a measured dependency shift, enabling rational experimental follow-up. DD is an association-based prioritization metric, not a causal test of synthetic lethality; all candidates require experimental validation. An open-source R package (paralogSL) and a reproducible pipeline are provided.

4. 关键词 (Keywords, 6 个, 3–10 范围内)

Synthetic lethality; paralog compensation; DepMap; CPTAC proteomics; Delta Dependency; dependency window

5. 作者与单位 (Authors)

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|--------|-------------|--------------|------------------------|---------|
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【待补】: Qingqing Mo 邮箱; 两位作者 ORCID (BMC 强烈建议通讯作者提供)。

6. 推荐审稿人 (Suggested Reviewers, 5 位)

| # | 姓名                    | 单位                            | 专长                                    | 邮箱   |
|---|-----------------------|-------------------------------|---------------------------------------|------|
| 1 | Dr. Francisca Vazquez | DepMap / Broad Institute      | CRISPR dependency                     | 【待补】 |
| 2 | Dr. Michael P. Snyder | Stanford University           | proteogenomics, CPTAC                 | 【待补】 |
| 3 | Dr. Jason Moffat      | University of Toronto         | synthetic lethality screening         | 【待补】 |
| 4 | Dr. Rameen Beroukhim  | Dana-Farber / Broad Institute | cancer genomics, paralog dependencies | 【待补】 |
| 5 | Dr. Bing Zhang        | Baylor College of Medicine    | CPTAC proteogenomics                  | 【待补】 |

Opposed reviewers: None

【待补】: BMC 系统要求推荐审稿人的机构邮箱 (不接受 Gmail 等私人邮箱), 投稿前请到各单位官网查到公开学术邮箱填入。

7. 上传文件清单 (Files)

| 系统文件类型            | 文件                                                                   | 说明               |
|-------------------|----------------------------------------------------------------------|------------------|
| Manuscript (PDF)  | manuscript.pdf                                                       | 29 页, 双倍行距 + 行号  |
| Manuscript source | 打包 zip: manuscript.tex + manuscript.bbl + output/figures/Fig1-4 .pdf | GB 要求可编辑 LaTeX 源 |

| 系统文件类型             | 文件                                                                                            | 说明                                                   |
|--------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------|
| Supplementary      | supplementary.pdf + 源打包<br>(supplementary.tex +<br>output/figures/FigS1-S11 .pdf,<br>无独立 bbl) | 17 页                                                 |
| Cover letter       | cover_letter.pdf (或将其纯文本粘贴到<br>系统 cover letter 框)                                             | 3 页                                                  |
| Graphical abstract | GraphicalAbstract.png                                                                         | 位于 output/figures/; caption 见<br>submission_texts.md |
| Highlights         | 在系统文本框逐条录入 (见第 10 节)                                                                          | 每条 ≤85 字符                                            |

LaTeX 打包命令 (在仓库根目录执行):

```
zip submission_src.zip manuscript.tex manuscript.bbl manuscript_refs.bib \
  output/figures/Fig1_Framework_Validation.pdf output/figures/Fig2_Proteomics.pdf \
  output/figures/Fig3_Clinical.pdf output/figures/Fig4_Translational.pdf
zip supplementary_src.zip supplementary.tex output/figures/FigS*.pdf
```

## 8. 声明勾选项 (Declarations, 逐项照抄)

| 系统字段                                       | 填写内容                                                                                                                                                                                                                                                                 |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethics approval and consent to participate | Not applicable. This study used exclusively publicly available, de-identified data.                                                                                                                                                                                  |
| Consent for publication                    | Not applicable.                                                                                                                                                                                                                                                      |
| Competing interests                        | The authors declare no competing interests.                                                                                                                                                                                                                          |
| Funding                                    | This research received no specific grant from any funding agency. (系统如有“ <b>No funding</b> ”复选框请勾选)                                                                                                                                                                  |
| Authors' contributions                     | Q.Q.M. and T.Z. conceived the study. T.Z. developed the methodology, performed all computational analyses, and developed the R package. Q.Q.M. curated clinical datasets and provided gynecological oncology domain expertise. Q.Q.M. and T.Z. wrote the manuscript. |
| Acknowledgements                           | We thank the DepMap, CPTAC, TCGA, PRISM, and cBioPortal teams for making their data publicly available. We also acknowledge the SynLethDB, HGNC, and UniProt teams for curated resources.                                                                            |

### Availability of data and materials (系统文本框):

All data are publicly available: DepMap 26Q1 from <https://depmap.org/portal/download/>; CPTAC proteomics from <https://cbioportal.org> and <https://pdc.cancer.gov>; TCGA PanCan from <https://xenabrowser.net>; PRISM from <https://depmap.org/portal/prism/>; HGNC gene families from <https://www.genenames.org/>. Processed datasets with cell line-level DD values and all supplementary tables are archived at Zenodo (<https://doi.org/10.5281/zenodo.21502030>). All analysis code is available under the MIT license: paralogSL R package v1.1.0 at <https://github.com/tjogzt/paralogSL> (<https://doi.org/10.5281/zenodo.21502113>); complete analysis pipeline at <https://github.com/tjogzt/paralog-sl-predictor> (<https://doi.org/10.5281/zenodo.21502030>). A single entry point (verify\_all.sh) recomputes all 165 audited numeric claims in under one minute.

9. 系统其他常见字段

| 字段                                | 填写内容                     |
|-----------------------------------|--------------------------|
| Related/preprint                  | 无预印本（如后续挂 bioRxiv 再更新）   |
| Previously submitted elsewhere    | No                       |
| All authors approve submission    | Yes (cover letter 已含此声明) |
| Not under consideration elsewhere | Yes (cover letter 已含此声明) |

10. Highlights（系统逐条录入，每条 ≤85 字符）

- 1. A single-subtraction metric (Delta Dependency) predicts paralog synthetic lethality
- 2. Interpretable composite (AUROC 0.831) matches best multi-feature classifier in CV
- 3. Protein-level paralog compensation is confirmed across seven CPTAC cohorts (n = 771)
- 4. Therapeutic window analysis nominates ARID1A-ARID1B as the leading SL candidate
- 5. Every quantitative claim is recomputable from raw data with a single command

Graphical abstract caption（较长，完整版见 submission\_texts.md）。

待办汇总（投稿前必须关闭）

- 1. 【注意】确定最终作者名单；补 Qingqing Mo 邮箱（第 5 节）
- 2. 【注意】补两位作者 ORCID（第 5 节）
- 3. 【注意】补 5 位推荐审稿人机构邮箱（第 6 节）
- 4. 【注意】摘要超限：采纳第 3 节压缩版（建议同步替换 manuscript.tex 并重编 PDF，保持文稿与系统一致）